

For telcos, IPTV is a promise of a wide range of revenue streams. When you consider the fact that IPTV is essentially an Internet line connected to your television, you can imagine it opening the doors to a gamut of Web services, be it data, video or voice. Telecommunication companies would like to become your sole communication link and your only source for both non-interactive and interactive entertainment. And they are betting on IPTV to take them there.

You will thus find phone companies jumping onto the IPTV bandwagon. Already, Reliance has announced its intention to roll out services leveraging this technology by the end of this year. Other telecom players such as Bharti (Airtel), BSNL and VSNL have also reported their plans to enter this segment.

How does it work?

If you have ever watched a video streamed over the Internet, you have already experienced IPTV in its broadest sense. Simply put, the technology leverages the Internet to bring you television. However, while the video you streamed over the Net was very likely a stuttering, pixelated, low-resolution Aishwarya Rai number, IPTV will bring you content as you expect and demand from TV—smooth, and at high resolution.

So how does it work? Let's stick to Aishwarya to understand that. Say your friendly neighbourhood telecom operator has licensed the rights to telecast the latest and the hottest item number featuring Ms Rai. First step at the telecom HQ would be to encode the raw video into a format more suited to streaming. While they could stick to MPEG-2, better compression would be afforded by a more up-to-date codec, such as H.264 or Microsoft's Windows Media format. (Incidentally, Reliance and Microsoft have signed deals to the effect that the former will use Microsoft TV as the software powering its IPTV offering.)

Right, now that we have a video clip in the required format, it is then broken down into IP packets to be compliant with the way the Inter-

net works. These packets are then multicast from a central HQ to local operators.

A multicast message is one that is transmitted to multiple recipients simultaneously—thus, while a broadcast message is sent to everyone, a multicast message is sent to specific groups; called multicast groups. The biggest benefit is that of bandwidth conservation.

To better understand the concept of multicasting, picture a tree: its trunk is the backbone of the entire structure, the one with the maximum bandwidth. The trunk then breaks into branches, and the branches then sprout leaves. Imagine data flowing through such a tree. A multicast stream would be transmitted only once from its trunk, through the major branches, and would then be distributed (multiplied, or technically, multiplexed) out to the leaves. A multicast data stream is thus a single stream until it reaches its endpoints. In the case of IPTV, each channel is multicast to several endpoints. Later on, we will see how this helps.

Internet guru Dave Clark describes the multicast model thus: "You put packets in at one end, and the network conspires to deliver them to anyone who asks [for them]". In our example, content is multicast to local operators, or from the network's perspective, to the local routers.

Before we head further, we would like to point out that the decoding of the Aishwarya video and its eventual streaming is a very one-dimensional view of what really transpires. The telco's HQ is a hive of network activity. This hub handles all sorts of data, including traditional voice content, video, high-quality music, e-mail packets from the Internet, etc. This hub-side view allows the HQ to guarantee a high level of quality over their streams. It also allows them to prioritise these streams to maximise the available bandwidth. Note that since each stream is multicast, there is no way for an end-point to request corrupted content again—to allow for this inflexibility, multicast streams transmit redundant packets. All this ensures that hassle-free and high-quality content is just a remote control away, without breaking the plug-and-play spell of our TV sets.

Back to our scheduled program: the video stream has reached the local operator. This is where all the boring and invisible, yet important, bits go in, the most important being the software controls. Here you would have elements controlling user authentication, user billing, channel change requests, video-on-demand requests, advertisements, and so on.

At a user's end, the connection goes into a set-top box, which arranges the packets and then decodes the incoming signal for your TV set. We said each channel was multicast to several endpoints, and that this was a good thing; now, note that the set-top box is acting as an end-point to the multicast content. Each channel is sent to a particular multicast group. Thus, when you change the channel, the set-top box simply changes its multicast group. This is as good as instantaneous, and thus, when it comes to IPTV, surfing through channels should be faster than ever.

The "last mile" to the user can be either over DSL twisted-pair or over cable. Rarely, this could also take place over optic cables, making for an

